For PIC programming:

User can using MPLAB to change the program, then using the following cable to download:



The above cable is pickit3 programmer, can you try to find at your country? If you cannot find it, and ask us to buy it in China, its price is 90USD, but DHL price is too expensive by DDP. Please think about it.

For USB interface work.

The USB interface works with the computer. But we did not supply the software, here we cannot supply software currently, because it's also under our development. But we can supply you the following "commumication protocol", user can program by themselves, and use the USB interface on control panel to work on XK-JS5B

Communication Protocol

1、Communication Protocol

The computer and the control panel of XK-JS5B are adopted the communication of master-slave mode, the computer is master station, the control panel is slave station, namely that the computer send the request instruction, then the control panel receive the request instruction and carry out the response according to the request.

1.1、READ one character

（1）The computer request message frame

Byte	Function	Numerical value	remark

		（decimal system）	
Byte0	Function Code	3	
Byte1	address		
Byte2		0	
Byte3		0	

（2）The control panel response message frame

Byte	Function	Numerical value （decimal system）	remark
Byte0	Function Code	3	
Byte1	address		
Byte2	Numerical value high 8 bit		
Byte3	Numerical value low 8 bit		

1.2、WRITE one character

（1）The computer request message frame

Byte	Function	Numerical value （decimal system）	remark
Byte0	Function Code	6	
Byte1	address		
Byte2	Numerical value high 8 bit		
Byte3	Numerical value low 8 bit		

（2） The control panel response message frame

Byte	Function	Numerical value （decimal system）	remark
Byte0	Function Code	6	
Byte1	address		
Byte2	Numerical value high 8 bit		
Byte3	Numerical value low 8 bit		

1.3、Communication parameter

9600,n,8,1

2、 The data that required communicated

address	Function	READ WRITE attribute
1	JOINT 1 speed setting	write
2	JOINT 1 position setting	write
3	JOINT 2 speed setting	write
4	JOINT 2 position setting	write
5	JOINT 3 speed setting	write
6	JOINT 3 position setting	write
7	JOINT 4 speed setting	write
8	JOINT 4 position setting	write
9	JOINT 5 speed setting	write
10	JOINT 5 position setting	write
11	JOINT 6 speed setting	write
12	JOINT 6 position setting	write
13	auto running: starting point JOINT 1 position setting	write
14	auto running: starting point JOINT 2 position setting	write
15	auto running: starting point JOINT 3 position setting	write
16	auto running: starting point JOINT 4 position setting	write
17	auto running: starting point JOINT 5 position setting	write
18	auto running: starting point JOINT 6 position setting	write
19	auto running: terminal point JOINT 1 position setting	write
20	auto running: terminal point JOINT 2 position setting	write
21	auto running: terminal point JOINT 3 position setting	write
22	auto running: terminal point JOINT 4 position setting	write
23	auto running: terminal point JOINT 5 position setting	write
24	auto running: terminal point	write

	JOINT 6 position setting	
25	Bit0：JOINT 1 manual operation button Bit1：JOINT 2 manual operation button Bit2：JOINT 3 manual operation button Bit3：JOINT 4 manual operation button Bit4：JOINT 5 manual operation button Bit5：JOINT 6 manual operation button Bit6： auto starting button Bit7： auto stop button Bit8： Bit9： Bit10： Bit11： Bit12： Bit13： Bit14： Bit15：	write
100	Bit0：Control mode（0-manual，1-auto） Bit1 ： JOINT1(Self-Body rotation) clockwise Bit2：JOINT1(Self-Body rotation) anticlockwise Bit3：JOINT2(lower arm swing) Forward Bit4：JOINT2(lower arm swing) Backward Bit5 ： JOINT3(upper arm swing)Upward Bit6：JOINT3(upper arm swing) Downward Bit7： JOINT4（wrist swing）upward Bit8： JOINT4（wrist swing）downward Bit9：JOINT5（the paw rotation）clockwise Bit10 ： JOINT5 （ the paw	read

	rotation）anticlockwise Bit11：JOINT6（the paw open & close）Loosen Bit12：JOINT6（the paw open & close）firmly grasp Bit13： Bit14： Bit15：	
101	JOINT1 current position	read
102	JOINT2 current position	read
103	JOINT3 current position	read
104	JOINT4 current position	read
105	JOINT5 current position	read
106	JOINT6 current position	read